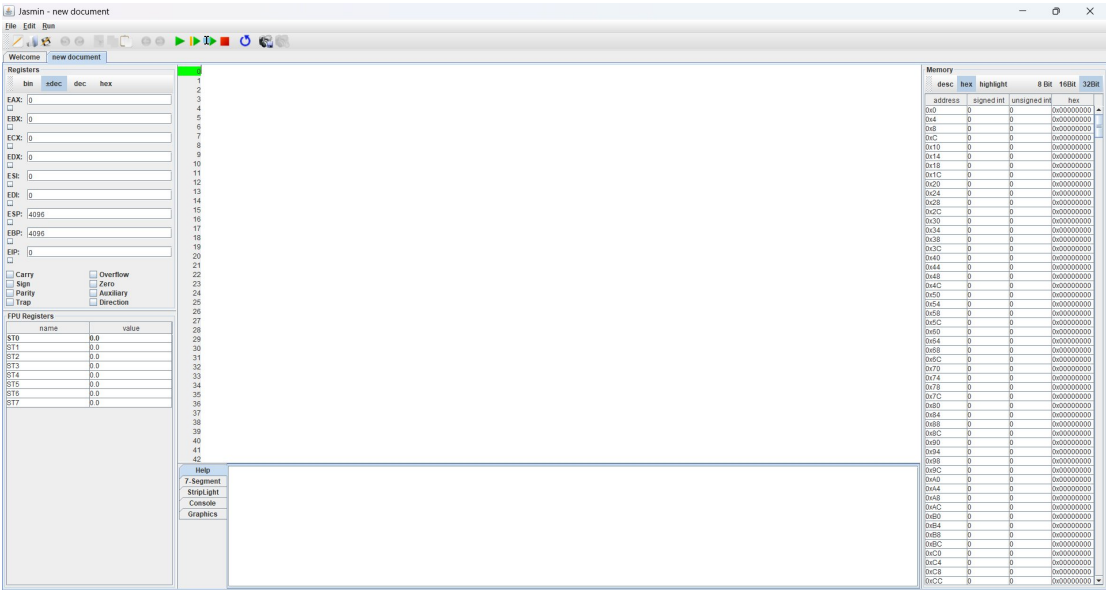


Nguyễn Trung Anh - HE190843
IA2001
FPT University

Lab 7

Using Jasmin to run x86 Assembly Code



Welcome new document

Registers

bin **±dec** dec hex

EAX: 4

EBX: 6

ECX: 0

EDX: 0

ESI: 0

EDI: 0

ESP: 4096

EBP: 4096

EIP: 2

☐ Carry ☐ Overflow
☐ Sign ☐ Zero

```

0 mov eax, 4
1 mov ebx, 6
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23

```

File Edit Run

Welcome new document

Registers

bin **±dec** dec hex

EAX: 4

EBX: 6

ECX: 10

EDX: 0

ESI: 0

EDI: 0

ESP: 4096

EBP: 4096

EIP: 6

☐ Carry ☐ Overflow
☐ Sign ☐ Zero
☒ Parity ☐ Auxiliary
☐ Trap ☐ Direction

```

0 mov eax, 4
1 mov ebx, 5
2 mov [eax], ebx
3 mov ecx, eax
4 add ecx, ebx
5 mov [eax+4], ecx

```

Memory

desc	hex	highlight	address	signed int	unsigned int	hex
0x0	0		0	0	0	0x00000000
0x4	6		4	6	6	0x00000006
0x8	10		8	10	10	0x0000000A
0xC	0		C	0	0	0x00000000
0x10	0		10	0	0	0x00000000
0x14	0		14	0	0	0x00000000
0x18	0		18	0	0	0x00000000
0x1C	0		1C	0	0	0x00000000
0x20	0		20	0	0	0x00000000
0x24	0		24	0	0	0x00000000
0x28	0		28	0	0	0x00000000
0x2C	0		2C	0	0	0x00000000
0x30	0		30	0	0	0x00000000
0x34	0		34	0	0	0x00000000
0x38	0		38	0	0	0x00000000
0x3C	0		3C	0	0	0x00000000
0x40	0		40	0	0	0x00000000
0x44	0		44	0	0	0x00000000
0x48	0		48	0	0	0x00000000
0x4C	0		4C	0	0	0x00000000
0x50	0		50	0	0	0x00000000
0x54	0		54	0	0	0x00000000
0x58	0		58	0	0	0x00000000

Registers

bin x86c dec hex

EAX: 1

EBC: 0

ECK: 19

EDK: 0

ESR: 0

EDR: 0

ESP: 4096

EBP: 4096

EP: 0

Carry Sign Parity Trap

Overflow Zero Auxiliary Direction

FPU Registers

name	value
ST0	0.0
ST1	0.0
ST2	0.0
ST3	0.0
ST4	0.0
ST5	0.0
ST6	0.0
ST7	0.0

7. Segment

String light

Console

Graphics

Help

Immediate value, general-purpose register, segment register or memory location

Usage: MOV DEST SRC

Effects: DEST ← SRC. Copies the source value (second argument) to the destination (first argument).

Flags to be set: none

Misc: Both arguments must be the same size (byte, word or doubleword). Move cannot be used to load the CS register, use JMP, CALL, or RET instead.

Memory

desc	hex	highlight	address	signed int	unsigned int	hex
0x0	0		0	0	0x00000000	
0x4	0		0	0	0x00000000	
0x8	0		0	0	0x00000000	
0x10	0		0	0	0x00000000	
0x14	0		0	0	0x00000000	
0x18	0		0	0	0x00000000	
0x1C	0		0	0	0x00000000	
0x20	0		0	0	0x00000000	
0x24	0		0	0	0x00000000	
0x28	0		0	0	0x00000000	
0x2C	0		0	0	0x00000000	
0x30	0		0	0	0x00000000	
0x34	0		0	0	0x00000000	
0x38	0		0	0	0x00000000	
0x3C	0		0	0	0x00000000	
0x40	0		0	0	0x00000000	
0x44	0		0	0	0x00000000	
0x48	0		0	0	0x00000000	
0x4C	0		0	0	0x00000000	
0x50	0		0	0	0x00000000	
0x54	0		0	0	0x00000000	
0x58	0		0	0	0x00000000	
0x5C	0		0	0	0x00000000	
0x60	0		0	0	0x00000000	
0x64	0		0	0	0x00000000	
0x68	0		0	0	0x00000000	
0x6C	0		0	0	0x00000000	
0x70	0		0	0	0x00000000	
0x74	0		0	0	0x00000000	
0x78	0		0	0	0x00000000	
0x7C	0		0	0	0x00000000	
0x80	0		0	0	0x00000000	
0x84	0		0	0	0x00000000	
0x88	0		0	0	0x00000000	
0x8C	0		0	0	0x00000000	
0x90	0		0	0	0x00000000	
0x94	0		0	0	0x00000000	
0x98	0		0	0	0x00000000	
0x9C	0		0	0	0x00000000	
0xA0	0		0	0	0x00000000	
0xA4	0		0	0	0x00000000	
0xA8	0		0	0	0x00000000	
0xAC	0		0	0	0x00000000	
0xB0	0		0	0	0x00000000	
0xB4	0		0	0	0x00000000	
0xB8	0		0	0	0x00000000	
0xBC	0		0	0	0x00000000	
0xC0	0		0	0	0x00000000	
0xC4	0		0	0	0x00000000	
0xC8	0		0	0	0x00000000	
0xCC	0		0	0	0x00000000	

Registers

bin x86c dec hex

EAX: 1

EBC: 0

ECK: 19

EDK: 0

ESR: 0

EDR: 0

ESP: 4096

EBP: 4096

EP: 0

Carry Sign Parity Trap

Overflow Zero Auxiliary Direction

FPU Registers

name	value
ST0	0.0
ST1	0.0
ST2	0.0
ST3	0.0
ST4	0.0
ST5	0.0
ST6	0.0
ST7	0.0

7. Segment

String light

Console

Graphics

Help

No help available for current context.

Memory

desc	hex	highlight	address	signed int	unsigned int	hex
0x30	0		0	0	0x00000000	
0x34	0		0	0	0x00000000	
0x38	0		0	0	0x00000000	
0x3C	0		0	0	0x00000000	
0x40	0		0	0	0x00000000	
0x44	0		0	0	0x00000000	
0x48	0		0	0	0x00000000	
0x4C	0		0	0	0x00000000	
0x50	0		0	0	0x00000000	
0x54	0		0	0	0x00000000	
0x58	0		0	0	0x00000000	
0x5C	0		0	0	0x00000000	
0x60	0		0	0	0x00000000	
0x64	0		0	0	0x00000000	
0x68	0		0	0	0x00000000	
0x6C	0		0	0	0x00000000	
0x70	0		0	0	0x00000000	
0x74	0		0	0	0x00000000	
0x78	0		0	0	0x00000000	
0x7C	0		0	0	0x00000000	
0x80	0		0	0	0x00000000	
0x84	0		0	0	0x00000000	
0x88	0		0	0	0x00000000	
0x8C	0		0	0	0x00000000	
0x90	0		0	0	0x00000000	
0x94	0		0	0	0x00000000	
0x98	0		0	0	0x00000000	
0x9C	0		0	0	0x00000000	
0xA0	0		0	0	0x00000000	
0xA4	0		0	0	0x00000000	
0xA8	0		0	0	0x00000000	
0xAC	0		0	0	0x00000000	
0xB0	0		0	0	0x00000000	
0xB4	0		0	0	0x00000000	
0xB8	0		0	0	0x00000000	
0xBC	0		0	0	0x00000000	
0xC0	0		0	0	0x00000000	
0xC4	0		0	0	0x00000000	
0xC8	0		0	0	0x00000000	
0xCC	0		0	0	0x00000000	
0xD0	0		0	0	0x00000000	
0xD4	0		0	0	0x00000000	
0xD8	0		0	0	0x00000000	
0xDC	0		0	0	0x00000000	
0xE0	0		0	0	0x00000000	
0xE4	0		0	0	0x00000000	
0xE8	0		0	0	0x00000000	
0xEC	0		0	0	0x00000000	
0xF0	0		0	0	0x00000000	
0xF4	0		0	0	0x00000000	
0xF8	0		0	0	0x00000000	
0xFC	0		0	0	0x00000000	

File Edit Run

Welcome | new document

Registers

bin sdec dec hex

EAX: 4

EBX: 6

ECX: 6

EDX: 0

ESI: 0

EDI: 0

ESP: 4092

EBP: 4096

EEP: 5

Carry Sign Parity Trap

Overflow Zero Auxiliary Direction

FPU Registers

name	value
ST0	0.0
ST1	0.0
ST2	0.0
ST3	0.0
ST4	0.0
ST5	0.0
ST6	0.0
ST7	0.0

asm

```
jmov eax, 4
jmov ebx, 6
jpush eax
jpush ebx
jpop ecx
```

Memory

desc	hex	highlight	address	signed int	unsigned int	hex	8 Bit	16Bit	32Bit
0xF30	0	0	0	0	0	0x00000000			
0xF34	0	0	0	0	0	0x00000000			
0xF38	0	0	0	0	0	0x00000000			
0xF3C	0	0	0	0	0	0x00000000			
0xF40	0	0	0	0	0	0x00000000			
0xF44	0	0	0	0	0	0x00000000			
0xF48	0	0	0	0	0	0x00000000			
0xF4C	0	0	0	0	0	0x00000000			
0xF50	0	0	0	0	0	0x00000000			
0xF54	0	0	0	0	0	0x00000000			
0xF58	0	0	0	0	0	0x00000000			
0xF5C	0	0	0	0	0	0x00000000			
0xF60	0	0	0	0	0	0x00000000			
0xF64	0	0	0	0	0	0x00000000			
0xF68	0	0	0	0	0	0x00000000			
0xF6C	0	0	0	0	0	0x00000000			
0xF70	0	0	0	0	0	0x00000000			
0xF74	0	0	0	0	0	0x00000000			
0xF78	0	0	0	0	0	0x00000000			
0xF7C	0	0	0	0	0	0x00000000			
0xF80	0	0	0	0	0	0x00000000			
0xF84	0	0	0	0	0	0x00000000			
0xF88	0	0	0	0	0	0x00000000			
0xF8C	0	0	0	0	0	0x00000000			
0xF90	0	0	0	0	0	0x00000000			
0xF94	0	0	0	0	0	0x00000000			
0xF98	0	0	0	0	0	0x00000000			
0xF9C	0	0	0	0	0	0x00000000			
0xFA0	0	0	0	0	0	0x00000000			
0xFA4	0	0	0	0	0	0x00000000			
0xFAB	0	0	0	0	0	0x00000000			
0xFAC	0	0	0	0	0	0x00000000			
0xFB0	0	0	0	0	0	0x00000000			
0xFB4	0	0	0	0	0	0x00000000			
0xFB8	0	0	0	0	0	0x00000000			
0xFBC	0	0	0	0	0	0x00000000			
0xFC0	0	0	0	0	0	0x00000000			
0xFC4	0	0	0	0	0	0x00000000			
0xFC8	0	0	0	0	0	0x00000000			
0xFCC	0	0	0	0	0	0x00000000			
0xFD0	0	0	0	0	0	0x00000000			
0xFD4	0	0	0	0	0	0x00000000			
0xFD8	0	0	0	0	0	0x00000000			
0xFDC	0	0	0	0	0	0x00000000			
0xFE0	0	0	0	0	0	0x00000000			
0xFE4	0	0	0	0	0	0x00000000			
0xFE8	0	0	0	0	0	0x00000000			
0xFEC	0	0	0	0	0	0x00000000			
0xFF8	6	6	6	6	6	0x00000006			
0xFFC	4	4	4	4	4	0x00000004			

Help

7-Segment

StripLight

Console

Graphics

Command: POP-Pop a Word from the Stack

Arguments: DEST: 16 or 32 bit memory location or register or segment register

Usage: POP DEST

Effects: POP stores the contents of DEST with the word on the top of the 80386 stack, addressed by SS:SP (address-size attribute of 16 bits) or SS:ESP (address-size attribute of 32 bits). The stack pointer is then increased accordingly

Flags to be set: none

Welcome | new document

Registers

bin sdec dec hex

EAX: 1

EBX: 2

ECX: 3

EDX: 4

ESI: 0

EDI: 0

ESP: 4096

EBP: 4096

EEP: 4

Carry Sign Parity Trap

Overflow Zero Auxiliary Direction

FPU Registers

name	value
ST0	0.0
ST1	0.0
ST2	0.0
ST3	0.0
ST4	0.0
ST5	0.0
ST6	0.0
ST7	0.0

asm

```
jmov eax, 1
jmov ebx, 2
jmov ecx, 3
jmov edx, 4
jpush eax
jpush ebx
jpush ecx
jpush edx
jpop eax
jpop ebx
jpop ecx
```

Memory

desc	hex	highlight	address	signed int	unsigned int	hex	8 Bit	16Bit	32Bit
0xF30	0	0	0	0	0	0x00000000			
0xF34	0	0	0	0	0	0x00000000			
0xF38	0	0	0	0	0	0x00000000			
0xF3C	0	0	0	0	0	0x00000000			
0xF40	0	0	0	0	0	0x00000000			
0xF44	0	0	0	0	0	0x00000000			
0xF48	0	0	0	0	0	0x00000000			
0xF4C	0	0	0	0	0	0x00000000			
0xF50	0	0	0	0	0	0x00000000			
0xF54	0	0	0	0	0	0x00000000			
0xF58	0	0	0	0	0	0x00000000			
0xF5C	0	0	0	0	0	0x00000000			
0xF60	0	0	0	0	0	0x00000000			
0xF64	0	0	0	0	0	0x00000000			
0xF68	0	0	0	0	0	0x00000000			
0xF6C	0	0	0	0	0	0x00000000			
0xF70	0	0	0	0	0	0x00000000			
0xF74	0	0	0	0	0	0x00000000			
0xF78	0	0	0	0	0	0x00000000			
0xF7C	0	0	0	0	0	0x00000000			
0xF80	0	0	0	0	0	0x00000000			
0xF84	0	0	0	0	0	0x00000000			
0xF88	0	0	0	0	0	0x00000000			
0xF8C	0	0	0	0	0	0x00000000			
0xF90	0	0	0	0	0	0x00000000			
0xF94	0	0	0	0	0	0x00000000			
0xF98	0	0	0	0	0	0x00000000			
0xF9C	0	0	0	0	0	0x00000000			
0xFA0	0	0	0	0	0	0x00000000			
0xFA4	0	0	0	0	0	0x00000000			
0xFAB	0	0	0	0	0	0x00000000			
0xFAC	0	0	0	0	0	0x00000000			
0xFB0	0	0	0	0	0	0x00000000			
0xFB4	0	0	0	0	0	0x00000000			
0xFB8	0	0	0	0	0	0x00000000			
0xFBC	0	0	0	0	0	0x00000000			
0xFC0	0	0	0	0	0	0x00000000			
0xFC4	0	0	0	0	0	0x00000000			
0xFC8	0	0	0	0	0	0x00000000			
0xFCC	0	0	0	0	0	0x00000000			
0xFD0	0	0	0	0	0	0x00000000			
0xFD4	0	0	0	0	0	0x00000000			
0xFD8	0	0	0	0	0	0x00000000			
0xFDC	0	0	0	0	0	0x00000000			
0xFE0	0	0	0	0	0	0x00000000			
0xFE4	0	0	0	0	0	0x00000000			
0xFE8	0	0	0	0	0	0x00000000			
0xFEC	0	0	0	0	0	0x00000000			

Help

7-Segment

StripLight

Console

Graphics

Command: POP-Pop a Word from the Stack

Arguments: DEST: 16 or 32 bit memory location or register or segment register

Usage: POP DEST

Effects: POP stores the contents of DEST with the word on the top of the 80386 stack, addressed by SS:SP (address-size attribute of 16 bits) or SS:ESP (address-size attribute of 32 bits). The stack pointer is then increased accordingly

File Edit Run

Welcome new document

Registers

bin **±dec** dec hex

EAX: 1
EBX: 2
ECX: 3
EDX: 4
ESI: 0
EDI: 0
ESP: 4080
EBP: 4095
EIP: 8

☐ Carry ☐ Overflow
☐ Sign ☐ Zero
☐ Parity ☐ Auxiliary
☐ Trap ☐ Direction

FPU Registers

name	value
ST0	0.0
ST1	0.0
ST2	0.0
ST3	0.0
ST4	0.0
ST5	0.0
ST6	0.0
ST7	0.0

0 mov eax, 1
1 mov ebx, 2
2 mov ecx, 3
3 mov edx, 4
4 push eax
5 push ebx
6 push ecx
7 push edx
8 pop eax
9 pop ebx
10 pop ecx
11 pop edx

Memory

desc hex highlight

address	signed int	unsigned int	hex
0xF44	0	0	0x00000000
0xF48	0	0	0x00000000
0xF4C	0	0	0x00000000
0xF50	0	0	0x00000000
0xF54	0	0	0x00000000
0xF58	0	0	0x00000000
0xF5C	0	0	0x00000000
0xF60	0	0	0x00000000
0xF64	0	0	0x00000000
0xF68	0	0	0x00000000
0xF6C	0	0	0x00000000
0xF70	0	0	0x00000000
0xF74	0	0	0x00000000
0xF78	0	0	0x00000000
0xF7C	0	0	0x00000000
0xF80	0	0	0x00000000
0xF84	0	0	0x00000000
0xF88	0	0	0x00000000
0xF8C	0	0	0x00000000
0xF90	0	0	0x00000000
0xF94	0	0	0x00000000
0xF98	0	0	0x00000000
0xF9C	0	0	0x00000000
0xFA0	0	0	0x00000000
0xFA4	0	0	0x00000000
0xFA8	0	0	0x00000000
0xFAC	0	0	0x00000000
0xFB0	0	0	0x00000000
0xFB4	0	0	0x00000000
0xFB8	0	0	0x00000000
0xFBC	0	0	0x00000000
0xFC0	0	0	0x00000000
0xFC4	0	0	0x00000000
0xFC8	0	0	0x00000000
0xFCC	0	0	0x00000000
0xFD0	0	0	0x00000000
0xFD4	0	0	0x00000000
0xFD8	0	0	0x00000000
0xFDC	0	0	0x00000000
0xFE0	0	0	0x00000000
0xFE4	0	0	0x00000000
0xFE8	0	0	0x00000000
0xFEC	0	0	0x00000000
0xFF0	4	4	0x00000004
0xFF4	3	3	0x00000003
0xFF8	2	2	0x00000002
0xFFC	1	1	0x00000001

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value (second argument) to the destination (first argument)
Flags to be set: none
Misc: Both arguments must be the same size (byte, word or doubleword). Move cannot be used to load the CS register, use JMP, CALL or RET instead.

File Edit Run

Welcome new document new document

Registers

bin **±dec** dec hex

EAX: 1
EBX: 2
ECX: 3
EDX: 4
ESI: 0
EDI: 0
ESP: 4095
EBP: 4095
EIP: 12

☐ Carry ☐ Overflow
☐ Sign ☐ Zero
☐ Parity ☐ Auxiliary
☐ Trap ☐ Direction

FPU Registers

name	value
ST0	0.0
ST1	0.0
ST2	0.0
ST3	0.0
ST4	0.0
ST5	0.0
ST6	0.0
ST7	0.0

0 mov eax, 1
1 mov ebx, 2
2 mov ecx, 3
3 mov edx, 4
4 push eax
5 push ebx
6 push ecx
7 push edx
8 pop eax
9 pop ebx
10 pop ecx
11 pop edx
12

Memory

desc hex highlight

address	signed int	unsigned int	hex
0xF30	0	0	0x00000000
0xF34	0	0	0x00000000
0xF38	0	0	0x00000000
0xF3C	0	0	0x00000000
0xF40	0	0	0x00000000
0xF44	0	0	0x00000000
0xF48	0	0	0x00000000
0xF4C	0	0	0x00000000
0xF50	0	0	0x00000000
0xF54	0	0	0x00000000
0xF58	0	0	0x00000000
0xF5C	0	0	0x00000000
0xF60	0	0	0x00000000
0xF64	0	0	0x00000000
0xF68	0	0	0x00000000
0xF6C	0	0	0x00000000
0xF70	0	0	0x00000000
0xF74	0	0	0x00000000
0xF78	0	0	0x00000000
0xF7C	0	0	0x00000000
0xF80	0	0	0x00000000
0xF84	0	0	0x00000000
0xF88	0	0	0x00000000
0xF8C	0	0	0x00000000
0xF90	0	0	0x00000000
0xF94	0	0	0x00000000
0xF98	0	0	0x00000000
0xF9C	0	0	0x00000000
0xFA0	0	0	0x00000000
0xFA4	0	0	0x00000000
0xFA8	0	0	0x00000000
0xFAC	0	0	0x00000000
0xFB0	0	0	0x00000000
0xFB4	0	0	0x00000000
0xFB8	0	0	0x00000000
0xFBC	0	0	0x00000000
0xFC0	0	0	0x00000000
0xFC4	0	0	0x00000000
0xFC8	0	0	0x00000000
0xFCC	0	0	0x00000000
0xFD0	0	0	0x00000000
0xFD4	0	0	0x00000000
0xFD8	0	0	0x00000000
0xFDC	0	0	0x00000000
0xFE0	0	0	0x00000000
0xFE4	0	0	0x00000000
0xFE8	0	0	0x00000000
0xFEC	0	0	0x00000000
0xFF0	4	4	0x00000004
0xFF4	3	3	0x00000003
0xFF8	2	2	0x00000002
0xFFC	1	1	0x00000001

Help
7-Segment
StripLight
Console
Graphics

Command: POP-POP a Word from the Stack
Arguments: DEST 16 or 32 bit memory location or register or segment register
Usage: POP DEST
Effects: POP stores the contents of DEST with the word on the top of the 80386 stack, addressed by SS:ESP (address-size attribute of 16 bits) or SS:ESP (address-size attribute of 32 bits). The stack pointer is then increased accordingly.
Flags to be set: none

IDA Pro

Finding the Address of DLLMain

Functions window		
Function name	Segment	Start
DllEntryPoint	.text	1001516D
wcslen	.text	1001520A
_Clacos	.text	10015210
_Clpow	.text	10015216
__dllonexit	.text	1001521C
type_info::~type_info(void)	.text	10015222

Find the import for gethostbyname

Imports			
Address	Ordinal	Name	Library
100163...		waveInStart	WINMM
100163C4	18	select	WS2_32
100163C8	11	inet_addr	WS2_32
100163...	52	gethostbyname	WS2_32
100163...	12	inet_ntoa	WS2_32
100163...	16	recv	WS2_32
100163...	19	send	WS2_32
100163...	4	connect	WS2_32
100163E0	15	ntohs	WS2_32
100163E4	9	htons	WS2_32
100163E8	21	setsockopt	WS2_32
100163...	116	WSACleanup	WS2_32
100163F0	115	WSAStartup	WS2_32

Count Local Variables for the Subroutine at 0x10001656


```

; DWORD __stdcall sub_10001656(LPVOID)
sub_10001656 proc near

var_675= byte ptr -675h
var_674= dword ptr -674h
hModule= dword ptr -670h
timeout= timeval ptr -66Ch
name= sockaddr ptr -664h
var_654= word ptr -654h
in= in_addr ptr -650h
Parameter= byte ptr -644h
CommandLine= byte ptr -63Fh
Data= byte ptr -638h
var_544= dword ptr -544h
var_50C= dword ptr -50Ch
var_500= dword ptr -500h
var_4FC= dword ptr -4FCh
readfds= fd_set ptr -4BCh
phkResult= HKEY__ ptr -3B8h
var_3B0= dword ptr -3B0h
var_1A4= dword ptr -1A4h
var_194= dword ptr -194h
WSAData= WSAData ptr -190h
arg_0= dword ptr 4

```

Finding the Purpose of the Code that References \cmd.exe /c

```

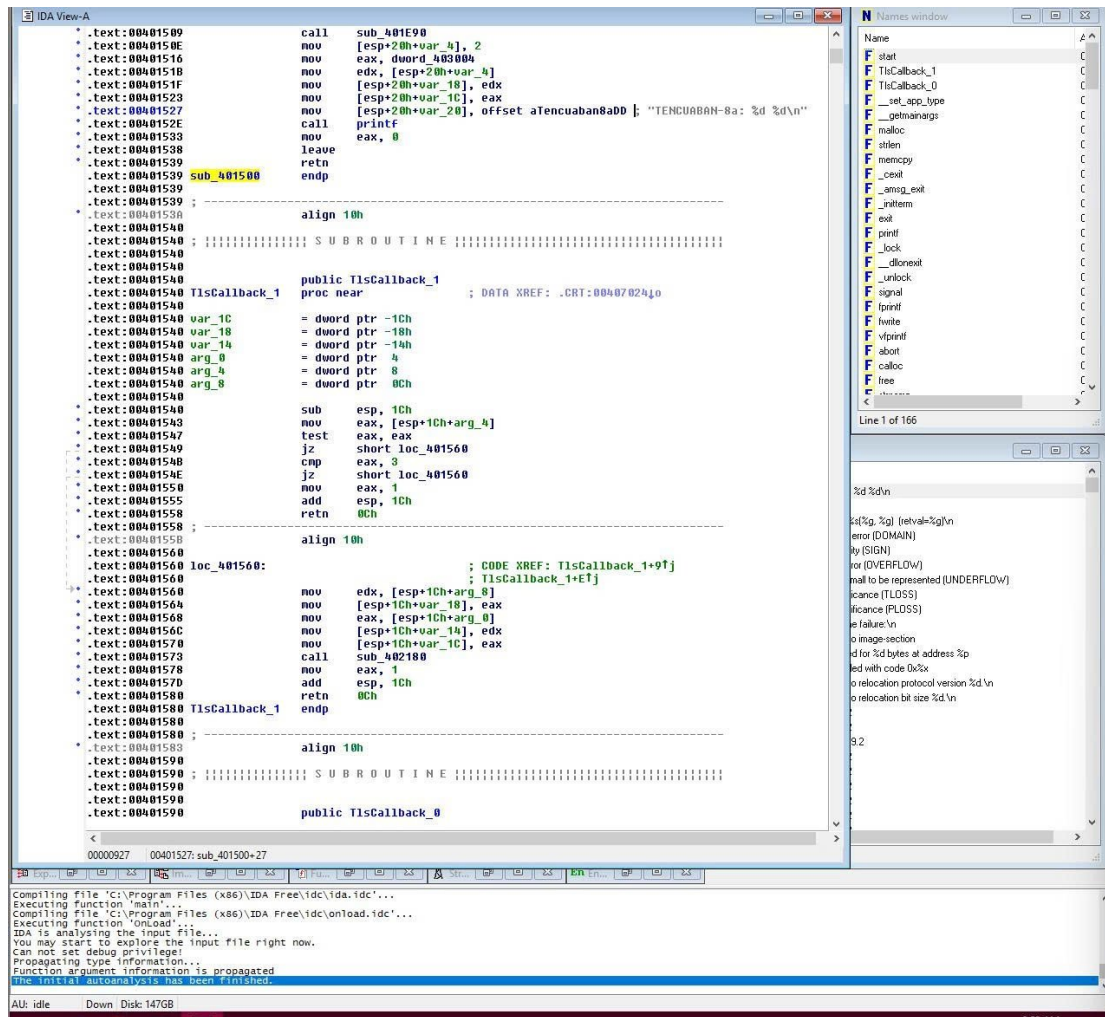
xdoors_d:10095B18 aQuit          db 'quit',0          ; DATA XREF: sub_1000FF58+36Fto
* xdoors_d:10095B1D                align 10h
* xdoors_d:10095B20 ; char aCommand_exeC[]
xdoors_d:10095B20 aCommand_exeC db '\command.exe /c ',0 ; DATA XREF: sub_1000FF58:loc_100101D7to
* xdoors_d:10095B31                align 4
* xdoors_d:10095B34 aCmd_exeC      db '\cmd.exe /c ',0      ; DATA XREF: sub_1000FF58+278to
* xdoors_d:10095B41                align 4
* xdoors_d:10095B44 ; char aHiMasterDDDDDD[]
xdoors_d:10095B44 aHiMasterDDDDDD db 'Hi,Master [%d/%d/%d %d:%d:%d]',00h,00h
* xdoors_d:10095B44                ; DATA XREF: sub_1000FF58+145to
xdoors_d:10095B44                db 'Welcome Back...Are You Enjoying Today?',00h,00h
xdoors_d:10095B44                db 00h,00h
xdoors_d:10095B44                db 'Machine UpTime [%-.2d Days %-.2d Hours %-.2d Minutes %-.2d Secon'
xdoors_d:10095B44                db 'ds]',00h,00h
xdoors_d:10095B44                db 'Machine IdleTime [%-.2d Days %-.2d Hours %-.2d Minutes %-.2d Seco'
xdoors_d:10095B44                db 'nds]',00h,00h
xdoors_d:10095B44                db 00h,00h
xdoors_d:10095B44                db 'Encrypt Magic Number For This Remote Shell Session [0x%02x]',00h,00h
xdoors_d:10095B44                db 00h,00h,0
xdoors_d:10095C5C ; char asc_10095C5C[]
xdoors_d:10095C5C asc_10095C5C:                ; DATA XREF: sub_1000FF58+4Bto
* xdoors_d:10095C5C                ; sub_1000FF58+3E1to
xdoors_d:10095C5C                dw 3Eh
xdoors_d:10095C5C                unicode 0, <>,0
* xdoors_d:10095C60                align 200h
* xdoors_d:10095C60 xdoors_d      ends
xdoors_d:10095C60
xdoors_d:10095C60

```

```
IDA View-A
* xdoors_d:10095B10 align 10h
* xdoors_d:10095B20 ; char aCommand_exeC[]
xdoors_d:10095B20 aCommand_exeC db '\command.exe /c ',0 ; DATA XREF: sub_1000FF58:loc_100101D7To
* xdoors_d:10095B30 align 4
xdoors_d:10095B30 aCmd_exeC db '\cmd.exe /c ',0 ; DATA XREF: sub_1000FF58+278To
* xdoors_d:10095B40 align 4
xdoors_d:10095B40 ; char aHiMaster000000[]
xdoors_d:10095B40 aHiMaster000000 db 'Hi,Master [%d/%d/%d %d:%d:%d]',00h,0Ah
xdoors_d:10095B40 ; DATA XREF: sub_1000FF58+145To
xdoors_d:10095B40 db 'Welcome Back...Are You Enjoying Today?',00h,0Ah
xdoors_d:10095B40 db 00h,0Ah
xdoors_d:10095B40 db 'Machine UpTime [%-.2d Days %-.2d Hours %-.2d Minutes %-.2d Secon'
xdoors_d:10095B40 db 'ds]',00h,0Ah
xdoors_d:10095B40 db 'Machine IdleTime [%-.2d Days %-.2d Hours %-.2d Minutes %-.2d Secon'
xdoors_d:10095B40 db 'nds]',00h,0Ah
xdoors_d:10095B40 db 00h,0Ah
xdoors_d:10095B40 db 'Encrypt Magic Number For This Remote Shell Session [0x%02x]',00h,0Ah
xdoors_d:10095B40 db 00h,0Ah,0
xdoors_d:10095C50 ; char asc_10095C5C[]
xdoors_d:10095C50 asc_10095C5C: ; DATA XREF: sub_1000FF58+h8To
xdoors_d:10095C50 ; sub_1000FF58+3E1To
* xdoors_d:10095C60 dw 3Eh
xdoors_d:10095C60 unicode 0, <,0
* xdoors_d:10095C60 align 200h
xdoors_d:10095C60 xdoors_d ends

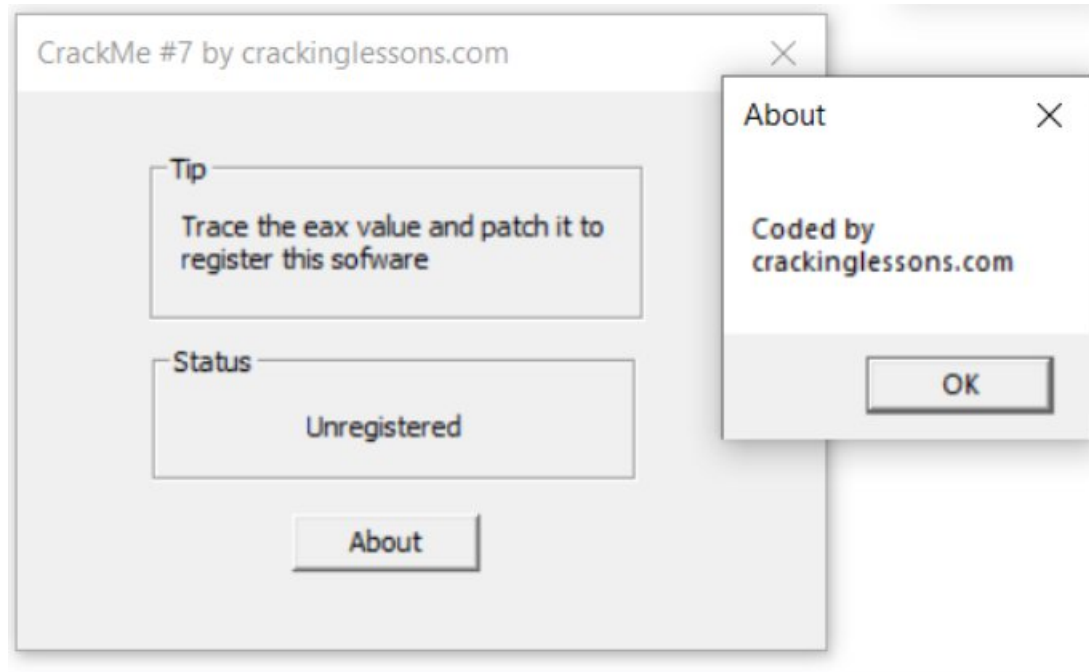
0001DF41 10095B41: xdoors_d:10095B41
```

Disassembling C on Windows



Bài tập Crackme 7

Lúc ban đầu



Em đưa file vào x32dbg và mở Strings References để tìm dòng chữ "Unregistered".

Em muốn xác định vị trí của nó trong code để có thể can thiệp.

0040101D	A3 A0424100	mov dword ptr ds:[4142A0],eax	eax:TpCallbackIndepend
00401022	B8 05000000	mov eax,5	eax:TpCallbackIndepend
00401027	B9 05000000	mov ecx,5	
0040102C	2BC1	sub eax,ecx	eax:TpCallbackIndepend
0040102E	85C0	test eax,eax	eax:TpCallbackIndepend
00401030	74 05	je crackme7.401037	
00401032	B8 02000000	mov eax,2	eax:TpCallbackIndepend
00401037	85C0	test eax,eax	eax:TpCallbackIndepend
00401039	74 07	je crackme7.401042	
0040103B	68 D81A4100	push crackme7.411AD8	411AD8:"Registered"
00401040	EB 05	jmp crackme7.401047	
00401042	68 E41A4100	push crackme7.411AE4	411AE4:"Unregistered"

test eax, eax rồi je – nếu eax = 0 thì nhảy đến "Unregistered".

Mà eax đang bằng 0 vì $5 - 5 = 0$.

Em sửa mov ecx,5 thành mov ecx,4 thì $eax = 5 - 4 = 1$ (khác 0), lúc đó je không nhảy nữa, chương trình chạy vào "Registered"

00401022	B8 05000000	mov eax,5	eax:TpCallbackIndepend
00401027	B9 05000000	mov ecx,5	eax:TpCallbackIndepend
0040102C	2BC1	sub eax,ecx	eax:TpCallbackIndepend
0040102E	85C0	test eax,eax	eax:TpCallbackIndepend
00401030	74 05	je crackme7.401037	
00401032	B8 02000000	mov eax,2	eax:TpCallbackIndepend
00401037	85C0	test eax,eax	eax:TpCallbackIndepend
00401039	74 07	je crackme7.401042	
0040103B	68 D81A4100	push crackme7.411AD8	411AD8:"Registered"
00401040	EB 05	jmp crackme7.401047	
00401042	68 E41A4100	push crackme7.411AE4	411AE4:"Unregistered"

Sau khi đổi nó như này

0040101D	A3 00724100	mov dword ptr [4176A0],eax	
00401022	B8 05000000	mov eax,5	
00401027	B9 04000000	mov ecx,4	
0040102C	2BC1	sub eax,ecx	
0040102E	85C0	test eax,eax	
00401030	74 05	je crackme7.401037	
00401032	B8 02000000	mov eax,2	
00401037	85C0	test eax,eax	
00401039	74 07	je crackme7.401042	
0040103B	68 D81A4100	push crackme7.411AD8	411AD8:"Registered"
00401040	EB 05	jmp crackme7.401047	
00401042	68 E41A4100	push crackme7.411AE4	411AE4:"Unregistered"
00401047	68 F0300000	push 300000F0	

DONE!

